

FIT5147 – Data Exploration and Visualisation

Monash University – Faculty of Information Technology

DEP Part 2: Data Exploration Project

Analysing Commuting Trends and Challenges in Melbourne's Rail Network

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1. Introduction

This project aims to investigate trends in passenger behaviour and recurring challenges within Melbourne’s metropolitan rail system. As a key component of the city’s public transport infrastructure, the rail network handles a substantial volume of daily commuter traffic. In recent years, issues such as peak-hour crowding and inconsistent service reliability have been frequently discussed in the media and government reports, particularly at major stations like Flinders Street and Southern Cross (Kolovos, 2023). Therefore, understanding how passenger volume changes across different times and identifying contributing factors is essential for improving train services and enhancing the overall commuter experience.

As a regular user of the network, I have often encountered delays and platform congestion during busy hours. These daily observations, combined with a personal interest in transport efficiency, have motivated me to examine the underlying patterns and causes through data analysis.

Following feedback from tutor Shivangi Gheewala during the proposal review stage, the original three research questions were revised. The third question, which focused on weather impacts, has been integrated as one of the external factors in the first question. Based on this revision, the project now addresses the following two core questions:

1. How do train service frequency and passenger numbers fluctuate across different time periods, and what are the internal and external factors (e.g., weather conditions) that influence these variations?
2. What methods can be used to quantify station crowding? How do factors (e.g., time of day, station design, service frequency) contribute to station crowding, and which stations are most affected?

Together, these two questions consist of four open-ended sub-questions and one relatively specific question. Through data-driven analysis, the project seeks to offer insights that may assist transport authorities, urban planners, and researchers in understanding passenger demand and improving service delivery and capacity planning.

2. Data Wrangling

2.1 Data Description

This project uses four datasets obtained from the Victorian Government open data portal. These datasets differ in structure, scale, and type, and together they provide a basis for temporal, spatial, and contextual analysis of Melbourne’s metropolitan rail network. The datasets are introduced below, along with their formats, sizes, content descriptions, and periods of data collection. For clarity, each dataset is assigned a shortened name for use in subsequent sections.

Train Service Passenger Counts (hereafter referred to as the *Passenger Dataset*)

Format: CSV | Size: 15.6 million rows × 21 columns | Period: July 2023 – June 2024

This tabular dataset records estimated passenger boardings and alightings at each station for Melbourne’s train services. Each row represents a single stop made by a train service at a station, along with associated passenger counts. The dataset includes variables such as date, scheduled time, station name, line name, travel direction, and total boardings/alightings. It also includes spatial coordinates for each station, allowing further integration with geographic data. [URL: Train Service Passenger Counts](#)

Argyle Square Weather Stations Historical Data (hereafter referred to as the *Weather Dataset*)

Format: CSV | Size: 138,538 rows | Period: May 2021 – July 2024

This dataset provides hourly weather observations recorded at Argyle Square in Melbourne. Each row corresponds to a single hour of weather data and includes variables such as temperature, precipitation, wind speed, and humidity. The data will be used to analyse external environmental factors that may affect train ridership. [URL: Argyle Square Weather Data](#)

Public Transport Stops (hereafter referred to as the *Station Dataset*)

Format: GeoJSON | Size: 28,323 rows | Date: November 2024

This spatial dataset provides location and metadata for public transport stops in Victoria, including Metro and Regional Train stops. Each row represents a stop or platform and includes fields such as stop ID, stop name, transport mode, and geographic coordinates. [URL: Public Transport Stops](#)

Public Transport Lines (hereafter referred to as the *Line Dataset*)

Format: GeoJSON | Size: 9,776 rows | Date: November 2024

This dataset provides geographic paths for public transport lines across Victoria, including rail routes. Each row describes a specific route variation, combining a direction and a unique trajectory. [URL: Public Transport Lines](#)

2.2 Preliminary Process

Before conducting data checking, several preliminary processing steps were applied to all four datasets. These steps included predefined column types during import, filtering and removal of irrelevant rows and columns, and unification of key column names to improve the efficiency and accuracy of subsequent data manipulation and merging.

First, column types were explicitly defined at the time of data import for each dataset. This approach improved import performance and ensured consistency in column types. For example, in the passenger dataset, day of week was defined as an ordered factor, while line name and direction were defined as unordered factors. The business date column was parsed as date type, and arrival time scheduled as time type. This type specification method was also applied to the other datasets.

Second, certain irrelevant rows and columns were removed. In the passenger dataset, only records where the mode column equals “Metro” were retained, and the mode column was subsequently dropped since other service types such as VLine are beyond the study scope. Similarly, records related to the Flemington Racecourse line were excluded, as it is a temporary event-based line. The “Group” column, which indicates the location group of each service, was also removed, as subsequent analysis will focus directly on specific train lines.

In the weather dataset, only records within the time range covered by the passenger dataset were kept. Columns unrelated to weather conditions, such as “dev_id”, “battery”, and “sensor_name”, were removed. For the station dataset and line dataset, records with “Mode” equal to “METRO TRAIN” were retained, and the “Mode” column was deleted, in order to keep the scope consistent across all datasets.

Finally, to facilitate merging between datasets, key column names used for joining were standardised. For example, “Station_Name” in the passenger dataset and “STOP_NAME” in the station dataset were renamed to a unified column name “Station_Name”. Similarly, “Line_Name” in the passenger dataset and “SHORT_NAME” in the line dataset were both renamed to “Line_Name”.

2.3 Data Checking and Cleaning

Before formal analysis, a unified data checking process was applied to all four datasets used in this project. The checking strategy is based on three common categories of data anomalies: coverage anomalies (e.g., missing or duplicate rows), syntactical anomalies (e.g., incorrect data types or formats), and semantic anomalies (e.g., logically invalid values). All checks were conducted in R. The specific methods are as follows:

- The *summary()* function was used to examine each column, check for missing values, detect invalid numerical entries, and observe distribution characteristics such as quartiles. Some of them, such as passenger-related columns, were checked using histogram.
- For ordered factor columns (e.g., day of week), *count()* was used to verify ordering levels and completeness.
- For unordered factor columns, if the number of levels was small, *summary()* was used to check for spelling inconsistencies; otherwise, *count()* was applied to review frequency distributions.
- For date, time, and datetime columns, standard formats (YYYY-MM-DD hh:mm:ss) were successfully applied in the preliminary stage, and time ranges were already aligned across datasets.
- For geographic coordinate columns, the *leaflet* package was used to render interactive maps for visual inspection of potential outliers, such as locations outside the Melbourne region or spatial duplications.

2.3.1 Passenger Dataset

Coverage anomalies: No missing values detected.

Syntactical anomalies: All column types were correctly assigned; no format issues observed.

Semantic anomalies: A very small number of records contained negative values in passenger-related columns. Since both boardings and alightings are explicitly recorded in the dataset, such negative values were considered invalid. Given the large overall data volume, the affected records were removed entirely. Some passenger-related columns contained extreme values; however, these were validated to correspond with busy stations or lines and were retained. The day type column included values such as “Normal Weekday,” “Saturday,” “Sunday,” “Public Holiday,” and “School Holiday.” “Saturday” and “Sunday” were grouped under “Normal Weekend” to reduce redundancy. No other semantic issues were identified.

2.3.2 Weather Dataset

Coverage anomalies: No missing values.

Syntactical anomalies: All columns are numeric with valid formats.

Semantic anomalies: The precipitation and strikes columns contained only zero values throughout. These were suspected to be sensor failures and were removed. Other weather variables, such as temperature, showed reasonable ranges with no extreme or unrealistic values.

2.3.3 Station Dataset

Coverage anomalies: No missing values, but near-duplicate records were found.

Syntactical anomalies: The geometry column was successfully read using *st_read()* with no structural issues.

Semantic anomalies: Multiple coordinates were associated with the same station in the leaflet map. For example, Flinders Street Station had 14 points. Upon inspection, these included main station entry, individual platforms, and bus replacement stop. Records with names containing “Railway” but not “Replacement” were extracted to represent valid train stations. The names were cleaned (removing “Railway”) to match other datasets. A new version of the Station Dataset was created, containing three columns: station name, number of platforms, and geometry. All coordinates were within the Melbourne region without obvious mismatch.

2.3.4 Line Dataset

Coverage anomalies: No missing values, but repeated geometries were found.

Syntactical anomalies: The geometry column was successfully imported; no format errors.

Semantic anomalies: Each train line included multiple trajectories (LineString geometries). For example, Werribee Line had 150 route LineString geometries. These routes represent different service paths but lack scheduling data, making it impossible to match directly with the passenger dataset. Since the line dataset is primarily used for plotting reference lines in [Figure 6](#), the longest trajectory (based on number of points inside the LineString geometry) was selected as the representative line for each service.

2.3.5 Summary

Overall, since all datasets originate from official government sources, the data quality is generally high, with few coverage or syntactical issues. Most identified anomalies were semantic in nature, including invalid values and redundant spatial entries. Some duplicate-like records were addressed as semantic issues rather than strict coverage anomalies.

2.4 Data Preparation

This section outlines how the cleaned data was further wrangled to support the visualisations in Section 3. Since each figure requires a different data structure, the preparation process is organised by figure. Unlike the general cleaning steps earlier, the operations here are task-specific and tailored to each visualisation, involving different subsets, aggregation methods, and time resolutions. Accordingly, the following subsections describe the preparation process for Figures 1 to 6.

Data Preparation for Figure 1

To measure the actual passenger volume in different directions for each metro line, we used the number of passengers boarding at each station as the primary metric. Passenger dataset was first grouped by train service line and direction, and the number of boardings was summed within each group to obtain the total number of passengers in both the forward and reverse directions for each line. The dataset was then reshaped into a wide format, and the total load (forward + reverse) was calculated and sorted to facilitate line ordering and annotation in the visualization. Click [here](#) to jump to Figure 1 lollipop chart.

Data Preparation for Figure 2

To compare the distribution of train frequency and passenger boardings across different train lines throughout the day, we first sorted the lines based on their total boarding volumes using a pre-computed summary table. Then, we constructed two separate datasets from passenger dataset: one representing train frequency (with each record assigned a weight of 1), and the other representing passenger boarding density (with each record weighted by the number of boardings). A new column, *Metric*, was added to distinguish between the two density types. The two datasets were then vertically concatenated to form a long-format data frame suitable for layered density plotting, in preparation for generating the [Figure 2](#) ridgeline plot.

Data Preparation for Figure 3

To identify the key factors influencing train passenger boardings, we first aggregated both the weather and passenger datasets at an hourly level (year-month-day-hour). The aggregated weather data includes hourly averages of meteorological variables such as solar radiation, wind speed, atmospheric pressure, and humidity. The passenger data was grouped to compute hourly train frequency (counting distinct train-station pairs), total passenger boardings, proportion of services in the “Towards Flinders” direction, day of the week, and day type (e.g., weekday, public holiday). These two datasets were then merged using the hourly level column via a left join, and unmatched rows with missing values were removed.

Considering the target variable, Total Passenger Boardings, follows a long-tailed distribution, we applied a logarithmic transformation to improve model performance. Before training, the dataset was randomly shuffled and split into training and test sets with an 80%-20% ratio. A random forest regressor was then trained, with tuning applied to key hyperparameters such as the number of trees (*n_{tree}*) and the number of features considered per tree (*m_{try}*). Based on the log-transformed target variable, we extracted two commonly used feature importance metrics: MSE Increase (increase in prediction error) and Node Purity Increase (improvement in node homogeneity). These two metrics were normalized using min-max scaling, averaged, and ranked to select the top 10 most important features for visualization ([Figure 3](#)).

Data Preparation for Figure 4

To visualize the hourly composition of station passenger volume across different day types, a new variable *Time_Bin* was created in passenger dataset based on the scheduled train arrival time, dividing the day into six 4-hour intervals. The data was then grouped by day type and time bin, and the average number of boardings and alightings was calculated for each group. These two values were summed to obtain the average station passenger volume, representing the typical level of foot traffic under specific temporal and calendar conditions.

To support the construction of pie charts, the proportion of boardings and alightings within each group was computed to determine the angular span of each pie segment. Meanwhile, the radius of each pie was determined by applying square-root scaling to the normalized average passenger volume. The resulting long form dataset includes the share of passenger flow types, average flow volume, and graphical position for each time–day combination, serving as the basis for [Figure 4](#).

Data Preparation for Figure 5 & 6

To quantify the level of crowding at each station, we developed a composite indicator called the Station Crowding Index (SCI), which is designed to incorporate multiple dimensions including time of day, day type, train frequency, passenger volume, and station infrastructure. The calculation of the SCI involves three main steps:

Step 1: Feature Extraction

From Passenger Dataset, we computed the following metrics:

- Average daily passenger volume at each station
- Average daily train count at each station
- Average passenger volume during 16:00–20:00 on normal weekdays at each station (this time period was identified as the peak from [Figure 4](#))

From the Station_Dataset, we obtained the number of platforms for each station.

Step 2: Deriving Four Component Dimensions

Using the extracted features, we defined the following four dimensions to characterize station crowding:

1. Peak Hour Volume: Average passenger volume during 16:00–20:00 on weekdays
2. Passenger Volume per Platform: Average daily passenger volume divided by the number of platforms
3. Train Count per Platform: Average daily train count divided by the number of platforms
4. Passenger Volume per Train: Average daily passenger volume divided by the average daily train count

Step 3: Normalization and Aggregation

Each of the four dimensions was scaled to a $[0, 1]$ range using min-max normalization, and their average was taken to compute the final SCI. A higher index value indicates a higher level of crowding pressure at the station. This index serves as the basis for the spatial visualizations presented in [Figure 5](#) and [Figure 6](#).

3. Data Exploration

3.1 Question 1

This section analyzes passenger numbers and train line frequency from the perspective of train lines, and presents three visualizations to answer [Question 1](#). The main datasets used in this section are the passenger dataset and the weather dataset.

Figure 1 shows the total number of passengers boarding on each metropolitan train line in both travel directions. Lines such as Pakenham, Lilydale, and Craigieburn record the highest overall boardings, while shorter or less busy lines like Alamein and Williamstown show comparatively lower totals. Directional differences are also apparent. For some lines, such as Pakenham and Cranbourne, the “Towards Flinders” direction accounts for a visibly larger share of the total volume; however, for Mernda, Hurstbridge, and Glen Waverley, the opposite is true. These comparisons help highlight not only the busiest lines but also directional imbalances in passenger flow.

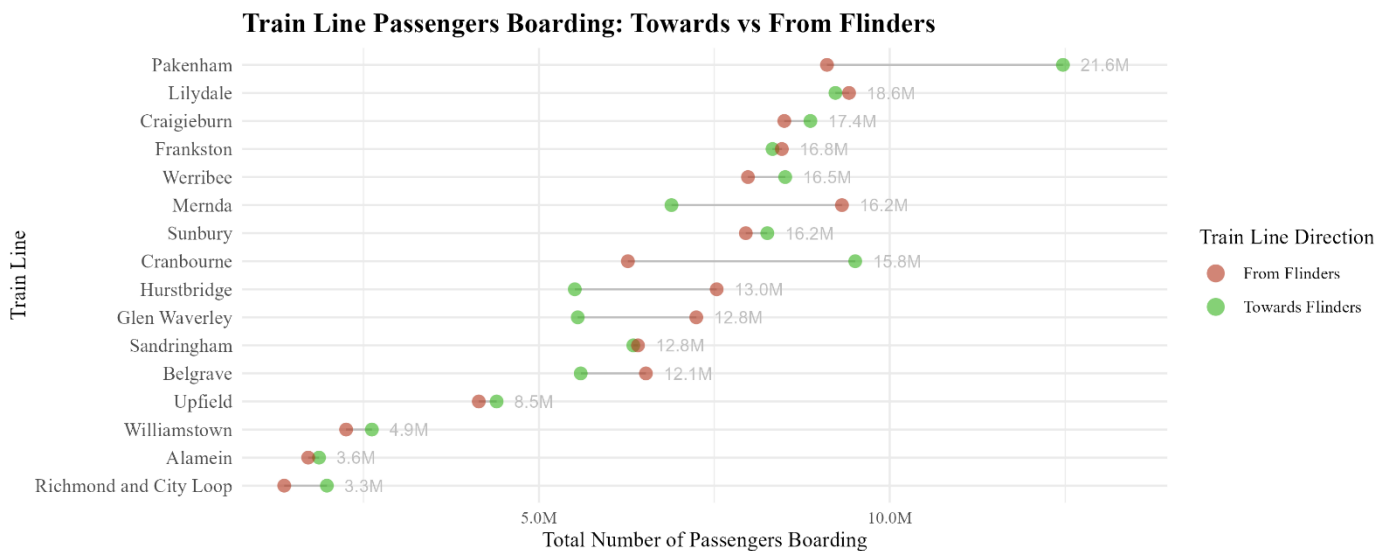


Figure 1: For some lines, opposite directions account for a visibly different share of the total volume.

Figure 1 is a lollipop chart, inspired by an example from the [ggplot2 graph gallery](#) (2025) and created using ggplot2 package. The chart uses position and length to represent passenger volumes, comparing two values—boarding counts in the “Towards Flinders” and “From Flinders” directions—for each train line. Two high contrast color hue is used as a categorical visual variable to distinguish direction: green for services towards Flinders Street Station and red for those away from it. The total passenger volume (sum of both directions) is labeled beside the greater value using grey text for emphasis without distraction.

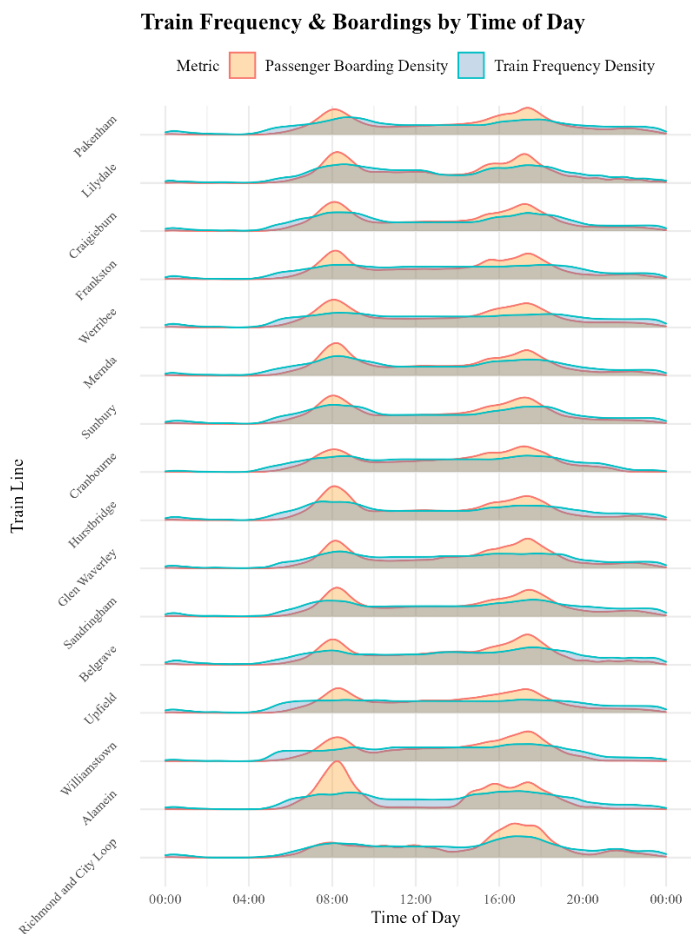


Figure 2: Supply and Demand

Figure 2 adopts a ridgeline plot format, inspired by [the ridgeline plot example in the ggplot2 gallery](#). Train lines are sorted by total passenger boardings. Color hues are used to distinguish metric types, with a visually contrasting blue-orange palette, and transparency is applied to emphasize overlapping areas. The x-axis represents time of day in hours, and the y-axis shows train line names. Key visual encodings include position (time), height (density), and color (metric type). This design communicates how well train operations align with passenger demand across different times of day.

Figure 3 presents the top 10 key features used to predict the hourly total passenger boardings. Each feature is evaluated using two different importance metrics from the random forest model: MSE Increase, which reflects the feature’s impact on the model’s prediction error, and Node Purity Increase, which measures its contribution to improving the purity of decision tree splits. Since the two metrics are on different scales and not directly comparable, both are normalized to a 0–1 range and visualized as grouped bar charts.

Figure 3 shows that train frequency stands out as the most important variable under both metrics, making it the most influential factor in predicting passenger boardings. It is followed by “Up Direction %” (proportion of trains heading inbound) and solar radiation. Notably, Solar Radiation is the only weather feature ranked in the top five, likely because it indirectly captures time of day, which is highly correlated with travel demand. Relative humidity is the second weather feature in the top 10. Its importance may come from its ability to indicate rainfall, which can influence travel decisions. Day of week and day type score relatively high on MSE Increase but low on Node Purity Increase, suggesting some inconsistency between the two

metrics. Conversely, features like solar radiation and “Up Direction %” rank high across both, indicating more consistent predictive power. Overall, internal features—such as train frequency and direction—play a dominant role in explaining passenger volumes, while weather factors contribute to a lesser extent.

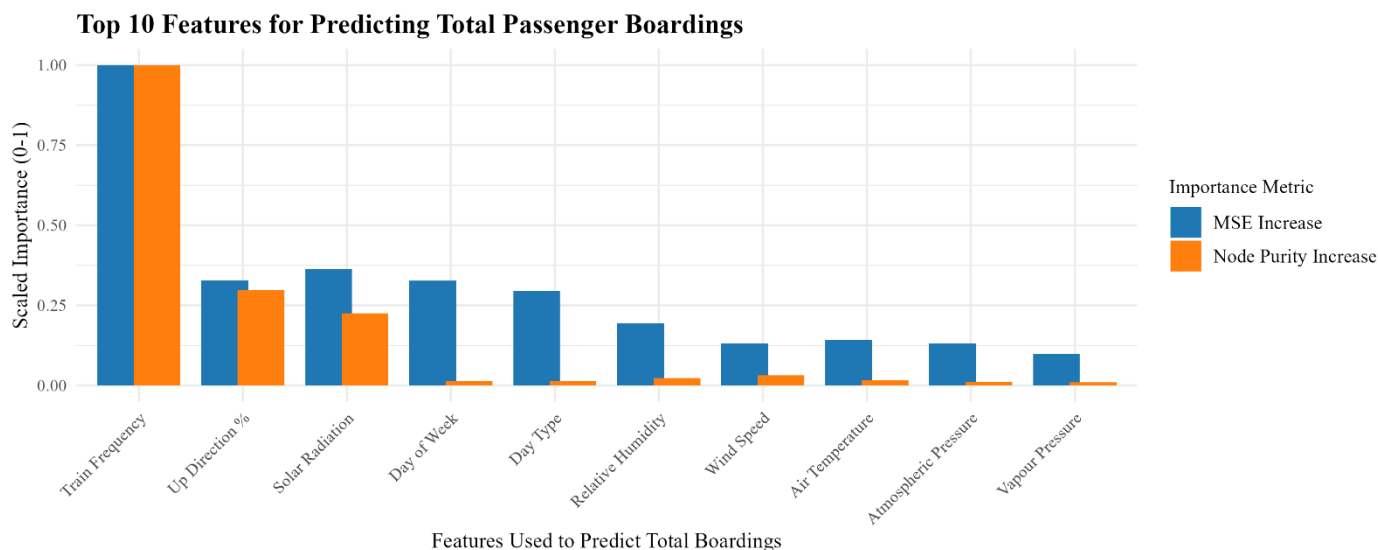


Figure 3: Weather factors contribute to a lesser extent.

Figure 3 is a grouped bar chart, with feature names on the x-axis, scaled importance on the y-axis, and color hue used to distinguish the two importance metrics. Blue represents MSE Increase and orange represents Node Purity Increase. Features are sorted by their average scaled importance to highlight the most influential ones. To avoid misinterpretation, numerical values are not displayed on top of each bar, as feature importance values are not directly interpretable and are intended only for relative comparison.

3.2 Question 2

This section focuses on measuring and analyzing station crowding from the perspective of train stations, using three visualizations to address [Question 2](#). This part draws on the passenger dataset, station dataset, and line dataset.

Figure 4 illustrates the composition of hourly station passenger volumes across different day types and time periods. Each pie chart corresponds to a unique combination of time bin and day type. The radius of each pie represents the average station passenger volume under that condition, while the angle of each slice indicates the proportion of boardings (blue) and alightings (orange). Two lines of text below each pie provide additional context: the boarding percentage and the average volume (with “pax” denoting passengers). Key observations through Figure 4 include:

- **By day type:** The average station passenger volume decreases from normal weekday to normal weekend to public holiday. Public holidays see about 60% of the weekday volume, while school holidays fall somewhere between weekdays and weekends.
- **By time of day:** The 16:00–20:00 time bin has the highest passenger volume, followed by 08:00–12:00. During 00:00–04:00, alightings significantly exceed boardings, whereas in 04:00–08:00, the reverse is true—boardings dominate. In other periods, the boarding-alighting proportions are relatively balanced.
- **Combined view:** The highest passenger activity occurs during 16:00–20:00 on weekdays, with an average of 41 passengers boarding or alighting per train per station. The lowest volume appears during 00:00–04:00 on public holidays, averaging only 4 passengers boarding or alighting per train per station.
- Notably, during the 04:00–08:00 time bin, passenger volumes differ significantly across day types. During this time bin on weekdays, the volume closely matches the 08:00–12:00 time bin, while during this time bin on weekends and public holidays it resembles 00:00–04:00. This suggests that peak hours start considerably later on non-working days.

Hourly Station Passenger Volume Composition by Day Type

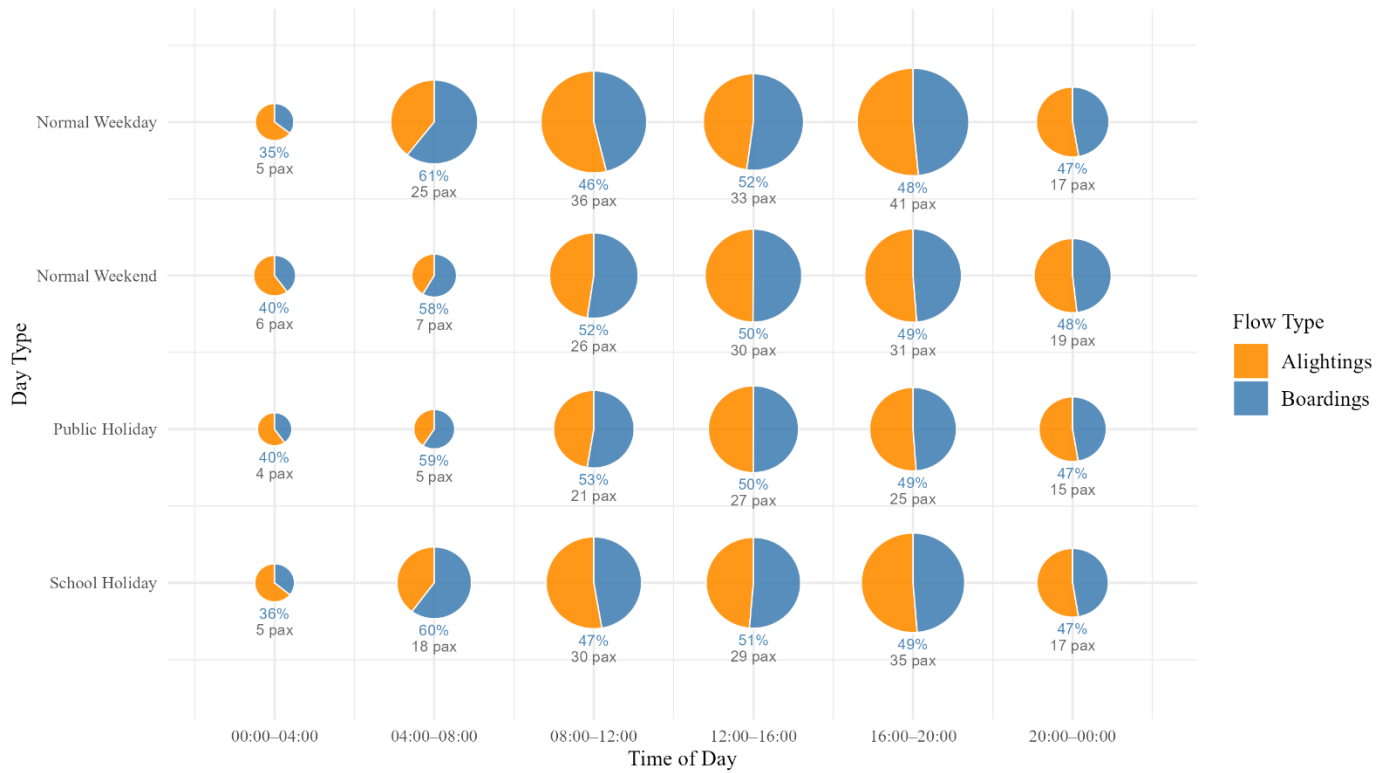


Figure 4: Peak hours start considerably later on non-working days.

Figure 4 adopts a faceted pie chart. To balance perceptual accuracy and visual clarity, pie radius is scaled using the square root of passenger volume. Two layers of annotations enhance readability by directly conveying boarding share and average passenger volume. This visualization utilizes multiple visual variables—position, area, angle, and color hue—to represent how station passenger flow patterns shift across time and day types. The chart is implemented using the `ggplot2` and `ggforce` packages.

Crowding Dimensions of the Top 3 Stations

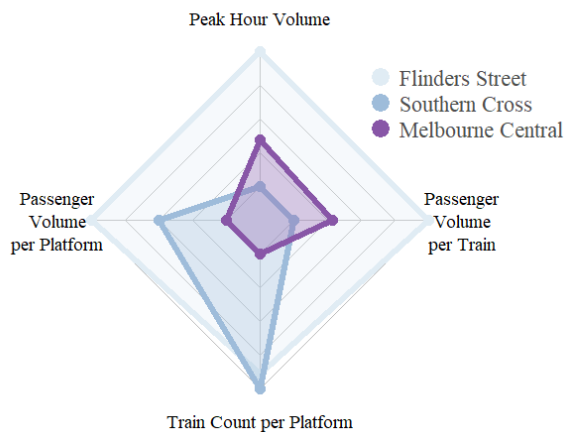


Figure 5: Top 3 Crowding Stations

Figure 5 illustrates the performance of the top three stations in terms of [Station Crowding Index \(SCI\)](#)—Flinders Street, Southern Cross, and Melbourne Central—across four normalized dimensions, aiming to reveal the different sources of crowding pressure at each station. Flinders Street scores close to the maximum in all four dimensions, with particularly higher values in peak hour volume and passenger volume per train than another two. Southern Cross shows a high value in train count per platform, but lower scores in passenger-related dimensions, suggesting its crowding is more driven by frequent train services. Melbourne Central demonstrates a more balanced but overall lower profile across all dimensions, making it the least crowded among the three.

The chart is a radar chart, designed to compare multiple variables across different entities. Each polygon represents a station, with values scaled (0–1) across the four dimensions. Different colors are used to distinguish stations, with semi-transparent fills and bold outlines to enhance readability. The design is inspired by [the radar chart example from the ggplot2 gallery](#) (2025), and was implemented using R packages such as `fmsb`.

Figure 6 presents the [Station Crowding Index \(SCI\)](#) across Melbourne’s metro network. This map shows that stations within the City Loop area have circles that are nearly black and visibly overlapping, highlighting their extremely high crowding. Along most lines, as the distance from the city loop increases, the circles become lighter in color and smaller in size, suggesting that stations farther from the city are generally less crowded. However, many terminal stations display slightly larger circles again, indicating a certain degree of crowding at line ends. The Cranbourne and Pakenham lines share a long overlapping segment where most stations exhibit orange-red circles, indicating notable crowding. This aligns with the finding in Figure 1, where the Pakenham line recorded the highest passenger volume. Similar patterns of overlapping crowding are observed on the shared segments of the Belgrave–Lilydale lines and the Williamstown–Werribee lines, making these three the busiest sections outside the City Loop. Among northern lines, the Craigieburn line shows relatively high crowding, indicating its importance in the north.

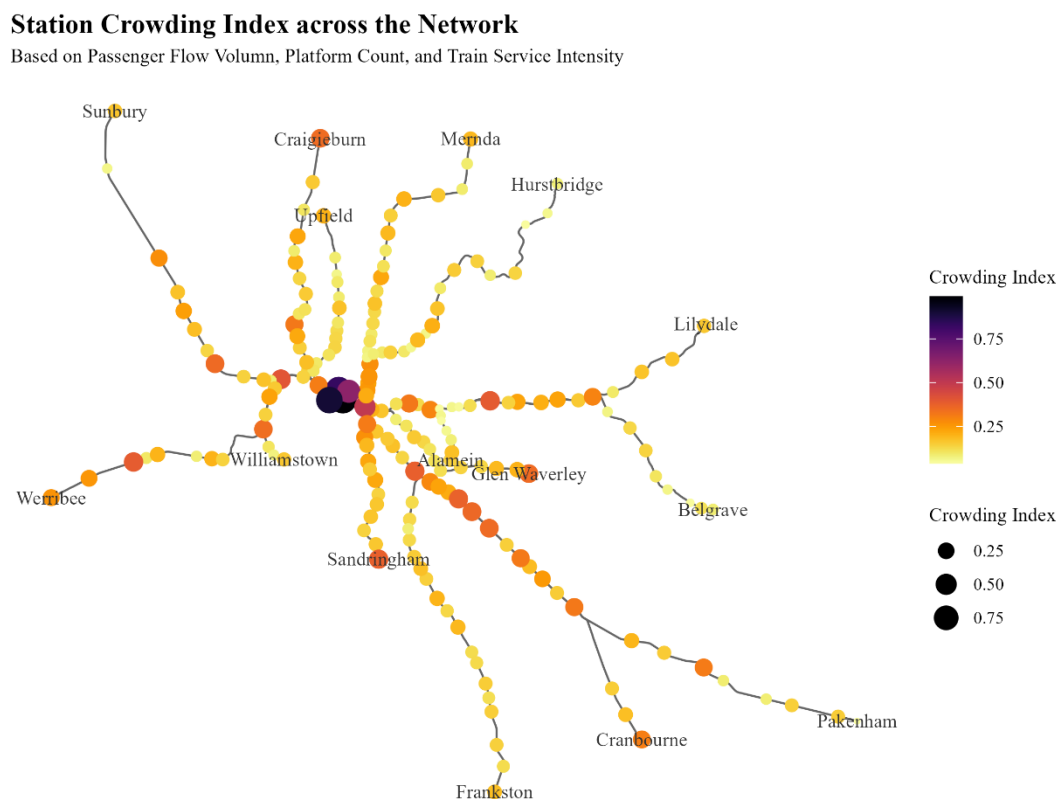


Figure 6: Stations within the City Loop area are extremely high crowding.

Figure 6 is a graduated symbol map overlaid with the train line reference paths. Position is determined by station coordinates, reflecting their geographic distribution. Color hue and value are encoded using the *viridis::inferno* gradient, ranging from yellow, orange, red, purple to black, representing increasing SCI values. Yellow indicates low crowding; black indicates the highest crowding levels. Circle size scales linearly with the SCI. The use of both color and size enhances the expressiveness and readability of crowding information. Train routes are connected with gray lines as a reference to network structure, and major lines are labeled at their termini.

4. Conclusion

Answer to Question 1: Variations in Passenger Volume and Train Frequency

This project first examined how passenger volume and train frequency varied across time on Melbourne’s metropolitan train lines. Results indicate that trunk lines such as Pakenham, Lilydale, and Craigieburn recorded the highest overall boardings and alightings, while branch lines like Alamein and Williamstown showed comparatively lower volumes. Moreover, directional imbalances were observed on several lines, with “towards the city” traffic significantly higher than another flow. Further analysis using density curve comparisons revealed a consistent pattern of demand exceeding supply during morning

and evening peak hours on most lines—passenger density surpassed train frequency, suggesting potential capacity shortfalls. In contrast, certain off-peak periods displayed excess supply. For example, the Alamein line had frequent midday services with low passenger demand, pointing to potential inefficiencies in scheduling. Among all lines, City Loop & Richmond exhibited the most balanced supply-demand alignment.

A feature importance analysis based on a random forest model identified train frequency as the most critical internal variable for predicting passenger volume, followed by the proportion of inbound lines and solar radiation. The relevance of solar radiation likely stems from its correlation with time of day, which strongly influences travel behavior. Weather features overall had limited impact, although relative humidity appeared to indirectly capture the influence of rainfall on travel decisions. In summary, internal operational factors play a dominant role in shaping passenger demand, while external factors like weather contribute less significantly to the observed patterns.

Answer to Question 2: Quantifying and Explaining Station Crowding

To assess station crowding, this project developed and applied a Station Crowding Index (SCI) to quantify the crowding level across different stations. The analysis revealed that crowding levels are jointly influenced by time of day, day type, and station location. First, comparisons across time periods and day types showed that weekdays between 16:00–20:00 experienced the highest passenger activity, while public holidays averaged only about 60% of weekday volumes. The 00:00–04:00 and 04:00–08:00 intervals exhibited distinct boarding-alighting ratios, reflecting typical travel behaviors at different hours. Notably, peak hour demand shifted significantly later on non-working days, suggesting a delayed start to travel patterns.

At the individual station level, Flinders Street exhibited the highest SCI, driven by high peak-hour passenger flow and boardings per train. Southern Cross showed high crowding primarily due to train frequency rather than passenger density. These differences indicate structural variation in crowding sources across stations.

Spatially, stations in the City Loop area experienced the most severe crowding, with SCI values approaching the upper limit and visually overlapping dark markers on the map. Along most train lines, SCI values decreased with distance from the city center, although some terminus stations showed renewed crowding. Shared segments such as Pakenham–Cranbourne and Belgrave–Lilydale also displayed higher crowding levels, consistent with their high passenger volumes. In summary, station crowding is primarily influenced by peak-period passenger density, train frequency, number of platforms, and spatial network structure.

5. Reflection

Throughout this project, I encountered several technical and conceptual challenges that helped deepen my understanding of data exploration in R. One major obstacle was the size of the passenger dataset, which was over 2.5 GB. This forced me to be extremely cautious with memory usage and processing efficiency throughout the workflow. Handling such a large dataset gave me an opportunity to strengthen my skills in *tidyverse*—I had not written over 1,500 lines of R code in a single project for quite some time. During the data wrangling phase, I finally grasped the practical importance of converting between wide and long table formats. For example, in creating Figure 4, I realized that a long format was necessary to produce the intended visual layout. This conceptual shift greatly improved my ability to restructure data for plotting. While I’ve already get in touch with *ggplot2*, I had never worked with the *sf* package before. Completing this project gave me a preliminary understanding of spatial data handling in R, although I still find the syntax and geometry concepts somewhat unfamiliar. I plan to revisit the official *sf* cheat sheet and experiment further. In hindsight, I initially hoped to create a map where train line widths and colors varied according to passenger volume. This concept, if realized, would likely offer a stronger visual impact than Figure 6. Unfortunately, despite a full day of effort, I was unable to correctly link LINESTRING geometry data with station-level passenger information. I plan to revisit this challenge in the Data Visualization Project (DVP). If there is a better approach to joining spatial line features with aggregated station-level metrics, I would be eager to learn it. Finally, I discovered the [R ggplot2 gallery](#) by chance during this project. It proved to be an invaluable resource for inspiration. Before producing each figure, I sketched a rough idea by hand, then browsed the gallery to refine the chart type or visual encoding. This process reminded me of the value of consulting documentation and learning from examples.

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